

18.06 Problem Set 12

Due 11am on Friday, December 2.

Problem 1 (5+5 points)

Suppose we are solving the ODE $\frac{dx}{dt} = Ax$ with a given initial condition $x(0)$, where A is an $m \times m$ matrix.

(a) Show that we can obtain the solutions $x(\Delta t), x(2\Delta t), x(3\Delta t), \dots$ at a sequence of discrete times $n\Delta t$ from $Bx(0), B^2x(0), B^3x(0), \dots$, i.e. by multiplying each preceding solution by some matrix B . (For example, we might use this to plot the solution at a sequence of points.) What is B ?

(b) We know from class that $x(t)$ must be **exponentially decaying** if all of the eigenvalues of A have **negative real parts**. $B^n x(0)$ is decaying if all of the eigenvalues of B have _____ — why must this follow if the eigenvalues of A have negative real parts?

Problem 2 (10+5 points)

Suppose that A is a 3×3 real-symmetric matrix. (Recall from class that such a matrix has real eigenvalues and orthogonal eigenvectors.) Suppose its eigenvalues are $\lambda_1 = 1, \lambda_2 = -1, \lambda_3 = -2$, and corresponding eigenvectors are x_1, x_2, x_3 . You are given that $x_1 = [1, 0, 1]$ (denoting a column vector ala Julia).

(a) Give an approximate solution at $t = 100$ to $\frac{dx}{dt} = Ax$ for $x(0) = [1, 1, 0]$. (Give a specific quantitative vector, even if the vector is very big or very small; an answer like " ≈ 0 " or " $\approx \infty$ " is not acceptable.)

(b) If $x_2 = [0, 1, 0]$, give a possible x_3 . (You should *not* use these x_2, x_3 to solve part (a).)

Problem 3 (3+3+5+3 points)

Let $X = (x_1 \ x_2 \ \cdots \ x_m)$ denote eigenvectors of the diagonalizable $m \times m$ **complex** matrix A , with corresponding eigenvalues $\lambda_1, \dots, \lambda_m$.

(a) The eigenvalues of A^H must be _____? Check your answer for a random complex matrix in Julia, computed with `A = rand(ComplexF64, 5, 5)`; note that A^H in Julia is `A'`.

(b) If A is real (so that $A^H = A^T$), why is your answer in (a) consistent with the statement in class that A and A^T have identical eigenvalues?

(c) From $A = X\Lambda X^{-1}$, derive a relationship between eigenvectors $y_1, \dots, y_m$ of A^H (the "left eigenvectors" of A) and the rows or columns of X^{-1} .

(d) Using the eigenvectors y_k from (c), what must be true of $y_1^H x_2$ (and similarly for other dot products)?

Problem 4 (5+5+2 points)

Suppose that A is an $m \times m$ **real-symmetric** matrix ($A = A^T$). Consider the function:

$$f(x) = \frac{x^T A x}{x^T x}$$

that take as input a real nonzero vector $x \in \mathbb{R}^m$ and returns a real number.

(a) Compute the gradient ∇f (with respect to x).

(b) Show that $\nabla f = 0$ if and only if x is some *eigenvector* of A , in which case $f(x)$ is equal to the eigenvalue!

(c) If the (real) eigenvalues of A are $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m$, then what are the minimum and maximum possible values of $f(x)$?